

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

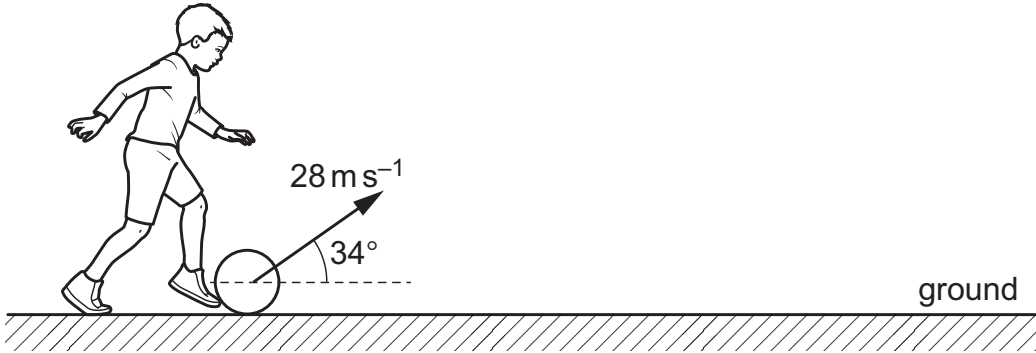


Fig. 1.1

Air resistance is negligible. The ball leaves the ground at time $t = 0$.

- (a) (i) Calculate the horizontal component v_H and the vertical component v_V of the velocity of the ball immediately after it has left the ground.

$v_H =$
.....
 m s^{-1}

$v_V =$
.....
 m s^{-1}

[2]

- (ii) Show that the ball reaches its maximum height at time $t = 1.6\text{ s}$.

[1]

- (iii) On Fig. 1.2, sketch the variation of v_H with time t between $t = 0$ and $t = 3.2\text{ s}$. Label your line H.

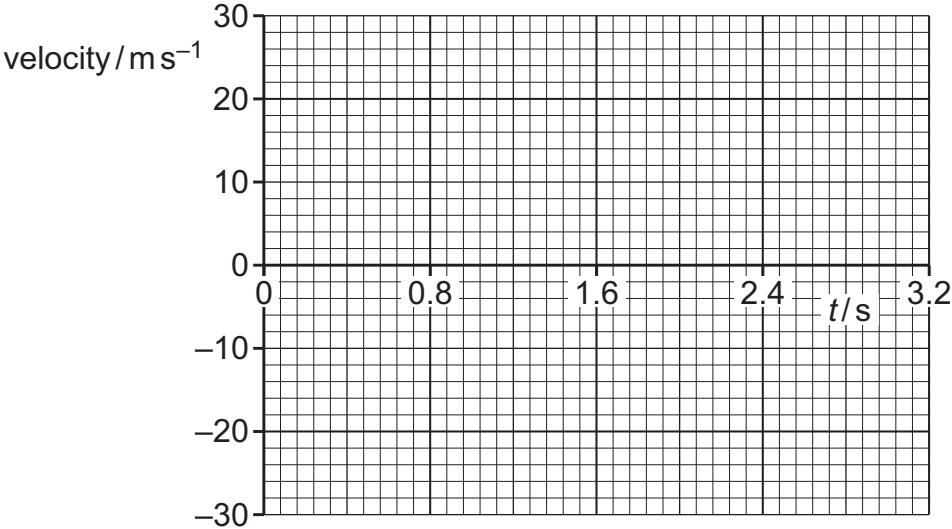


Fig. 1.2

[1]

- (iv) On Fig. 1.2, sketch the variation of v_V with time t between $t = 0$ and $t = 3.2\text{ s}$. Assume that velocity in the upward direction is positive. Label your line V.

[3]

- (b) The total change in momentum of the ball between leaving the ground at $t = 0$ and landing on the ground at $t = 3.2\text{ s}$ is 13 kg m s^{-1} .

- (i) Define momentum.

.....

.....

[1]

- (ii) Calculate the force that acts on the ball while it is in the air.

force =

.....

N[2]

- (iii) Determine the mass of the ball.

mass =

.....

kg [1]